

Chapter 60. Surfactant

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Surfactant: Introduction

Pulmonary surfactant lines the inner layer of the lung and serves to lower surface tension at the air–liquid interface, thereby maintaining alveolar stability. In the absence of surfactant, the work of breathing increases markedly ultimately resulting in respiratory failure secondary to atelectasis, alveolar flooding, and severe hypoxemia. The clinical correlate of surfactant deficiency is the neonatal respiratory distress syndrome (NRDS) in preterm infants, which before the mid-1980s was a devastating and fatal disease. Since the advent of exogenous surfactant replacement therapy, however, newborn mortality from NRDS has decreased approximately 56% from 1987 to 1995.¹ The primary surfactant deficiency of NRDS is now a well-characterized condition, and does not appear as complex as the various surfactant changes occurring during acute lung injury (ALI) and/or the acute respiratory distress syndrome (ARDS). Alterations of the endogenous surfactant system in the mature lungs of patients with these disorders are not as well understood, but currently represent an area of intense investigation. This complexity has resulted in inconsistent results of clinical trials evaluating exogenous surfactant administration in this patient population.

This chapter reviews surfactant metabolism and function in the mature lung and its role in maintaining normal lung homeostasis, including its more recently described host defense functions. The metabolism and function of surfactant in the injured lung is also outlined, with particular reference to the effects of mechanical ventilation on the alveolar surfactant system. Subsequently, the status of clinical trials evaluating exogenous surfactant administration in patients with ALI or ARDS is addressed, as are the various factors that may influence a host's response to this therapy. Future research directions relevant to the understanding of the role of surfactant both in ALI and ARDS and other lung diseases conclude the chapter.

Surfactant Composition and Metabolism

Composition

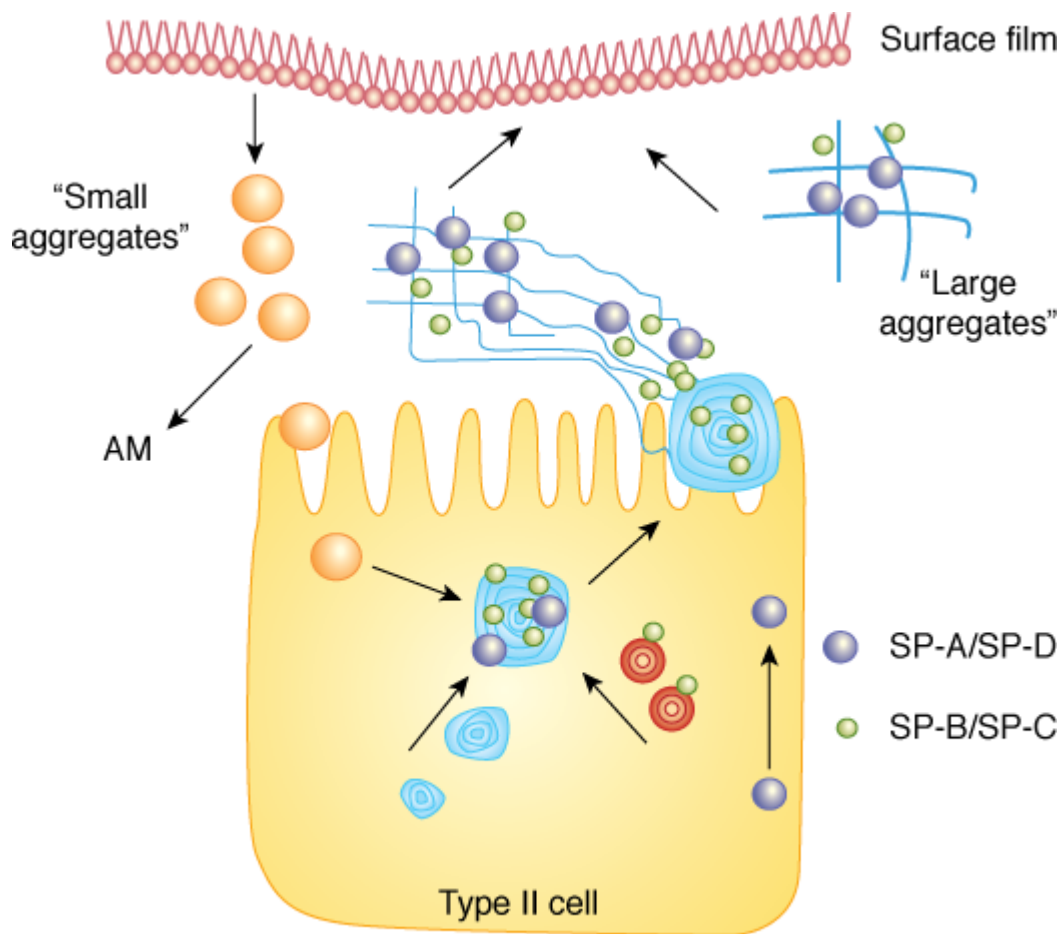
The composition of surfactant is remarkably similar among mammalian species, consisting of approximately 80% phospholipids, 10% surfactant-associated proteins,^{2–4} and approximately 10% neutral lipids, among which cholesterol is the most abundant (80% to 90% by weight).^{5,6} The major phospholipid component is phosphatidylcholine, half of which is the disaturated species, dipalmitoylphosphatidylcholine (DPPC).⁴ This

latter molecule is the major surface-active component responsible for lowering surface tension at the air–liquid interface, and is an essential component of all exogenous surfactant preparations currently available for clinical use. Other lipids include phosphatidylglycerol and a few minor lipid species, which are thought to be important in the generation and maintenance of the surface film. The surfactant-associated proteins have been designated as SP-A, SP-B, SP-C, and SP-D.^{7,8} SP-B and SP-C are small, hydrophobic proteins closely associated with the phospholipids, where they play a major role in generating and maintaining the surface tension-reducing surface film.^{9–11} SP-B is an 18-kDa dimer, whereas SP-C is a 4-kDa monomer, the latter being the more hydrophobic of these two proteins.^{12,13} The most clinically effective exogenous surfactant preparations currently in use contain at least one of these natural hydrophobic proteins, or similar types of synthetic and/or recombinant molecules. SP-A is an octodecamer made up of six trimers in a “bouquet” arrangement.^{14,15} Under reducing conditions, it is a 28-kDa monomer with a 35-kDa glycosylated form. SP-D is a large, multimeric cruciform structure, which is 42 kDa under reducing conditions. Both proteins are very hydrophilic and belong to the collectin family of proteins. They are not components of any of the available natural exogenous preparations consequent to their removal with purification processes, and are not yet available as synthetic or recombinant molecules. Recent evidence suggests that they play a more important role in host defense than in biophysical functions, so there has been renewed interest in developing exogenous preparations containing some form of these proteins.^{14–17}

Intracellular Metabolism

Surfactant is synthesized within alveolar type II cells (Fig. 60-1).¹⁸ Initial assembly occurs in the endoplasmic reticulum with intracellular transport via the Golgi apparatus. Surfactant is stored within lamellar bodies of the type II cell and is secreted into the airspace via exocytosis.^{19,20} Studies that have investigated the intracellular metabolic pathways of surfactant lipids and proteins have used radiolabeled precursors injected both intravenously and intratracheally. Basically, these studies have shown that although the hydrophobic proteins are assembled and secreted in conjunction with the lipids, SP-A and D are metabolized separately.²¹ For example, the dominant route for SP-A secretion is by direct, constitutive pathways independent of lamellar body exocytosis, although smaller amounts undergo regulated secretion in association with these organelles.^{22–24} SP-D is also metabolized independently of surfactant lipids, and, unlike the other surfactant proteins, has been identified in nonpulmonary organs such as the gut.^{25,26} Within the healthy lung, various pharmacologic agents, such as β -agonists can stimulate surfactant secretion, as can physical stretch of alveolar type II cells.²⁷ This latter phenomenon is particularly relevant to specific aspects of mechanical ventilation because higher tidal volumes increase the stretch of the alveolus, leading to immediate surfactant secretion from type II cells. These effects are relatively short-lived, however, and may be quite different within the injured lung, where the preexisting health of type II cells, before the onset of mechanical ventilation, may be compromised.

Figure 60-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Surfactant metabolism starts with synthesis within the type II cell, secretion into the airspace as lamellar bodies containing both lipid and protein components, and formation of tubular myelin structures. Large aggregates are composed of tubular myelin and freshly secreted lamellar bodies and represent precursors to the surface film. With respiratory motion, small aggregates are formed, which are taken back up into type II cells or cleared via alveolar macrophages (AM).

Recent studies have focused on the extracellular metabolism of pulmonary surfactant, once the material has been secreted from the type II cell into the alveolus (see Fig. 60-1). Within the airspace, mechanical ventilation has a significant and immediate impact on the surfactant system by influencing the metabolism of extracellular surfactant aggregate forms. Because this effect may have important clinical consequences in patients with ALI or ARDS, and is directly affected by different modes of mechanical ventilation, a more detailed discussion of this area is warranted.

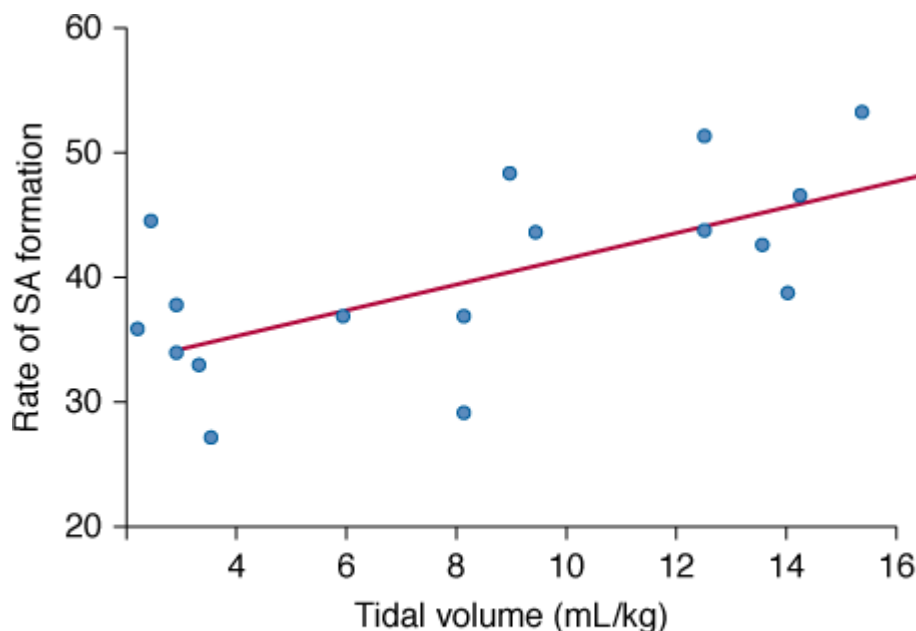
Extracellular Metabolism

Once secreted into the airspace, alveolar surfactant undergoes physical rearrangement from the lamellar structures into tubular myelin, a process involving SP-A, SP-B, DPPC, phosphatidylglycerol, and calcium.^{28,29} After differential centrifugation of isolated lung lavage, large lipid structures containing SP-A, SP-B, and SP-C, representing the heavier and functionally active forms of alveolar surfactant form a pellet,

and are called large aggregates (LA).^{30,31} They adsorb rapidly to the air–liquid interface and are subsequently converted into smaller, vesicular forms called small aggregates (SA) (see Fig. 60-1). The SA subfractions are poorly functioning, contain little surfactant protein, and are thought to represent surfactant forms that have left the surface film and are subsequently available for reuptake via type II cells for resynthesis into new LA, and/or are cleared entirely from the airspace via catabolism, mainly within alveolar macrophages.^{32–34}

The process of conversion of the functionally active LA forms into inactive SA within the airspace is specifically relevant when discussing the effects of mechanical ventilation on surfactant metabolism. In vitro studies using the surface area cycling technique show that this conversion of LA into SA is mediated by two main factors: a carboxylesterase enzyme called *convertase* and a phasic change in surface area.^{31,35–37} The phasic change in surface area factor has been investigated in vivo using different tidal volumes to mechanically ventilate normal rabbits. These studies show that increasing tidal volume, but not positive end-expiratory pressure, results in an increased conversion of LA into SA, and that this conversion occurs relatively soon after the onset of ventilation (Fig. 60-2).³⁸ In normal lungs, these changes do not result in significant alterations in aggregate pool sizes, presumably because of the capabilities of the normal alveolar environment to regulate surfactant metabolism. This does not appear to be the case in the injured lung, however, where alterations in surfactant metabolism occur both intracellularly and extracellularly.^{39,40} For example, within the lungs of patients with ALI or ARDS, increased aggregate conversion associated with higher tidal volumes is thought to result in increased SA pools, which, together with the associated decrease in functionally active LA pools, may contribute to progressive lung dysfunction. Indeed, these specific changes have been demonstrated in animal models of lung injury and have not only underscored the importance of instituting and maintaining optimal modes of mechanical ventilation within injured lungs, but also in the setting of exogenous surfactant, both during and after administration. This latter situation is discussed in more detail in subsequent sections.

Figure 60-2



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The rate of small aggregate (SA) formation increases in direct proportion to the tidal volumes used to ventilate normal adult rabbits. SA formation was determined by measuring ^3H -label recovery in the SA fraction of surfactant 1 hour after injection of a trace dose of ^3H -labeled large aggregates into the animals' lungs. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Ito Y, Veldhuizen RA, Yao LJ, et al. Ventilation strategies affect surfactant aggregate conversion in acute lung injury. *Am J Respir Crit Care Med*. 1997;155(2):493–499. Official Journal of the American Thoracic Society.)

Surfactant Physiology in the Normal Lung

Biophysical Function

The major role of the surfactant system is related to its biophysical function of lowering surface tension at the air–liquid interface.^{3,4,41} This is most evident clinically in preterm infants born deficient in surfactant who quickly die of respiratory failure unless supplemented with an exogenous surfactant preparation.^{42–44} As noted previously, the major lipid component contributing to lowering surface tension is DPPC, although the hydrophobic proteins and some other lipids are also essential.^{10–13} Full-term babies born deficient in SP-B develop severe lung dysfunction resistant to traditional surfactant therapy and ultimately require lung transplantation for survival.⁴⁵ Likewise, mice deficient in this protein suffering a similar outcome have also been shown to have abnormal processing of SP-C.⁴⁶ Mice deficient in SP-C alone, on the other hand, have relatively minor biophysical abnormalities, which predominantly manifest only at low lung volumes, suggesting a potential role for this protein in the surface film stabilization.⁴⁷

From a physiologic perspective, the initial discovery that surface tension was important for lung stability was made by von Neergaard in 1929, when he observed that it took more pressure to inflate air-filled than saline-filled lungs.⁴⁸ This was the direct result of the surface tension forces existing at the air-liquid interface. This concept is further advanced by the Laplace law, which states that the pressure gradient across a sphere, which we can roughly extrapolate to an alveolus, is directly related to the surface tension within the sphere divided by the radius of the sphere ($\Delta P = 2\delta/r$), where P is pressure, δ is surface tension, and r is radius. From this equation, it is clear that during exhalation, when the alveolar radius decreases, the tendency for this alveolus to collapse increases (i.e., ΔP) unless surface tension decreases as well. Because of its strategic location at the air-liquid interface, and the presence of both hydrophobic and hydrophilic regions, a surfactant film lining the air-liquid interface is able to lower surface tension to near zero levels as the film is compressed at low lung volumes. This decrease in surface tension not only serves to maintain alveolar stability and decrease the work of breathing, but also prevents alveolar flooding because high surface tension tends to draw fluid from the interstitium into the airspace.⁴⁹ Finally, because surfactant also lines the conducting airways, its biophysical function serves to maintain the patency and stability of small airways and enhance ciliary clearance of particles.^{50,51} These latter observations suggest that surfactant function may also be relevant in diseases such as asthma, chronic bronchitis, and cystic fibrosis.

Host-Defense Function

As noted previously, the surfactant proteins SP-A and SP-D are members of the collectin family and, secondary to their molecular composition and structure, can bind a variety of bacteria, fungi, allergens, and viruses, thus mediating phagocytosis by immune cells and the killing of pathogens within the lung.^{14,16,52} In addition, these proteins, as well as some of the hydrophobic components of surfactant, can influence production of [nitric oxide](#), [oxygen](#) radicals, and inflammatory mediators from activated cells.⁵³⁻⁵⁶ Underscoring the importance of SP-A in host-defense functions are *in vivo* studies demonstrating that mice deficient in this protein (which are phenotypically normal when not stressed) had increased pulmonary inflammation compared to wild-type animals when bacteria, viruses, or lipopolysaccharide was instilled into their lungs.^{57,58} These changes were mitigated when exogenous SP-A was administered to these animals. Similar, albeit slightly modified functions of SP-D have been shown both *in vitro* and *in vivo*; unlike the normal phenotype of the SP-A null mice, however, the phenotype of unstressed SP-D knockout animals is abnormal with enlarged airspaces and significantly altered surfactant pool sizes.^{59,60}

Although critical for the spreading and stabilization of the surfactant film, SP-B and SP-C also seem to contribute to the host-defense properties of surfactant. Mice deficient in SP-C show greater susceptibility to respiratory syncytial virus infection⁶¹ and have increased pulmonary inflammation, decreased macrophage phagocytic activity, and decreased survival compared to wild-type animals following intratracheal instillation of bacteria.⁶² The role of SP-B in host defense appears to be related to the aminoterminal propeptide of the protein, which improves phagocytosis and has antimicrobial activity at acidic pH.⁶³

The host-defense functions of the various surfactant lipids have also been evaluated. It has been shown that these phospholipids downregulate inflammation and [nitric oxide](#) production, presumably because of their “coating” effects on particles and cells.^{17,64–66} Additionally, surfactant lipids, and DPPC in particular, could have a direct effect in down-modulating the inflammatory response of immune cells, through the regulation of specific molecular pathways responsible for reactive [oxygen](#) species production.⁶⁷ Given the emerging role of the various surfactant components in host defense, future studies involving exogenous surfactant administration may focus more on inflammatory outcomes in addition to biophysical and physiological benefits.

Surfactant Physiology in the Injured Lung

Surfactant Alterations in Acute Lung Injury and Acute Respiratory Distress Syndrome

The definitions of ALI and ARDS are rather simplistic and are based on physiologic criteria more than specific etiologies. For example, any acute inflammatory insult that ultimately affects the lungs causing decreased lung compliance, hypoxemia, and bilateral infiltrates on chest radiography (not of cardiac origin) essentially fulfills the diagnostic criteria of ALI and ARDS.^{68,69} The pathophysiologic changes that occur during this process, including the surfactant alterations reported in numerous studies, are complex and reflect the lack of effective treatments for this disorder.

The first postmortem descriptions of the lungs of patients dying from ARDS suggested that surfactant dysfunction may have played a role in their demise.⁷⁰ Subsequently, many studies have reported consistent changes in the surfactant system in lavage samples obtained from these patients, including decreased phosphatidylcholine and DPPC levels, decreased phosphatidylglycerol levels, decreased SP-A, SP-B, SP-C, and SP-D levels, and a decrease in the functionally active LA forms relative to SA.^{71–73} Interestingly, in these severely ill patients, serum levels of SP-A and SP-D were shown to be elevated, likely consequent to the associated marked increase in pulmonary permeability in this setting.^{74,75} The biophysical consequence of these collective changes in the endogenous surfactant system is an inability to adequately reduce surface tension, resulting in decreased lung compliance, increased permeability with edema formation, and hypoxemia. The next section addresses the mechanisms responsible for the surfactant changes observed in these patients. Only with a better understanding of these mechanisms will optimal treatment strategies be developed that are aimed at restoring normal surface tension forces within injured lungs, or perhaps, more importantly, preventing the development of these changes.

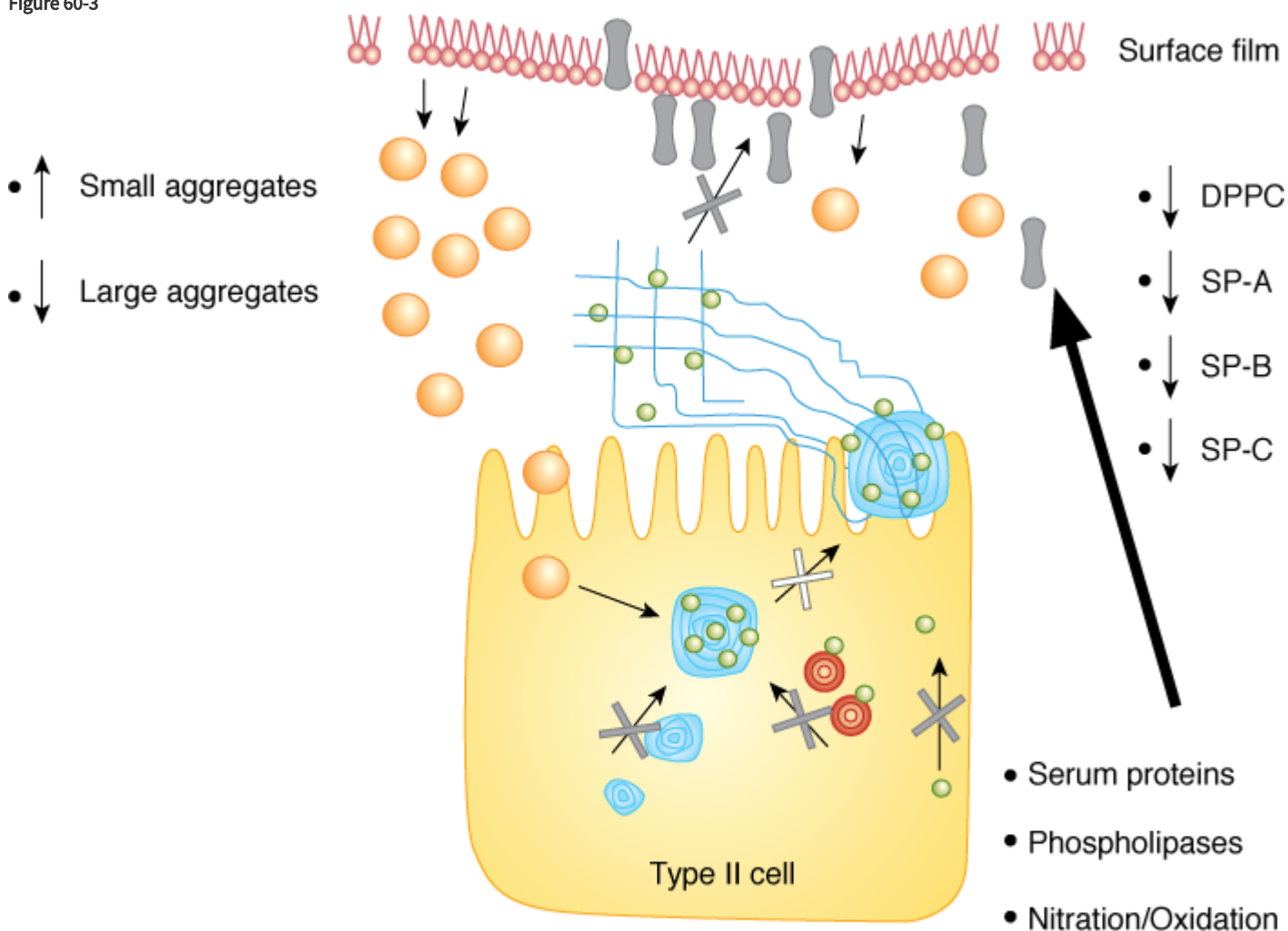
Mechanisms Responsible for Surfactant Alterations

Unlike NRDS, in which the preterm animal represents a reasonable correlate of the clinical condition of NRDS and is thus suitable for reliable mechanistic and therapeutic studies, no one animal model adequately reflects the complexity of patients with ALI or ARDS. Proposed mechanisms in the latter setting are,

therefore, derived from many different animal models, a factor that needs to be considered when extrapolating results of such studies to the clinical situation.

The observed changes in the phospholipid composition of endogenous surfactant within the injured lung represent a relatively late finding in models of ALI when severe lung dysfunction is present. These changes are likely related to abnormal synthetic and/or secretory pathways within the type II cell, as well as alterations in the degradation process of some lipids within the airspace via increased phospholipase activity (Fig. 60-3).^{76–79} A similar mechanism is likely responsible for the decreased surfactant protein levels observed in these lungs, although with the permeability abnormalities demonstrated in these patients, there may also be transfer of SP-A, SP-B, and SP-D across the alveolar–capillary barrier into the serum.^{74,75} Interestingly, clinical studies show that serum levels of SP-A and SP-B are inversely related to oxygenation values in patients with ARDS, and serum SP-D levels positively correlate with the survival of these patients.^{74,75} In addition to these quantitative changes in surfactant protein levels, there may also be modifications to their composition in the form of nitration and/or oxidation, which would also compromise their function but may not impact measurable levels.^{80–82}

Figure 60-3



Mechanisms leading to surfactant alterations in the injured lung include abnormal type II cell metabolism, increased phospholipase activity, nitration and oxidation of surfactant, increased conversion of large aggregates into small aggregates, and inhibition of surfactant by serum proteins leaking into the airspace. See text for details.

The mechanisms responsible for the changes observed in surfactant aggregate forms were alluded to earlier. Briefly, decreased LA pools may be caused by decreased type II cell synthesis and/or secretion in the severely injured lung, but an increased conversion of LA into SA forms likely occurs at earlier stages of the disease, particularly in patients undergoing mechanical ventilation.^{37–40} This was recently investigated in animal studies in which spontaneously breathing septic rats with relatively mild lung dysfunction had unchanged LA pools, whereas SA pools were significantly decreased compared to sham animals.⁸³ These changes were opposite to those documented in severely injured lungs; it was hypothesized that they may have represented the host's compensatory response aimed at maintaining endogenous surfactant in LA forms. For example, by utilizing smaller tidal volumes (with higher respiratory rates), these animals would decrease LA conversion secondary to the smaller changes in alveolar surface area, thereby attempting to maintain lung function. Indeed, very soon after the onset of mechanical ventilation of these animals, which involved using higher tidal volumes than those associated with spontaneous breathing, conversion of LA into SA increased dramatically and lung function deteriorated rapidly.⁸⁴ Of clinical relevance, however, was the ability to mitigate both the aggregate conversion and progressive lung dysfunction observed by using lower tidal volumes from the onset of ventilation. In fact, high-frequency oscillation (HFO), a strategy involving extremely small tidal volumes with high respiratory rates, proved to be superior to all other “lung-protective” modes of ventilation with respect to aggregate conversion, lung function, and inflammatory changes.⁸⁵ Interestingly, these laboratory results are consistent with recent clinical trials evaluating different modes of mechanical ventilation in patients with ALI. One large trial showed that lower tidal volumes resulted in superior outcomes compared to higher tidal volumes in patients with ALI or ARDS.⁸⁶ Results of phase III clinical trials evaluating the outcome of patients specifically ventilated with HFO are pending.

Recent studies suggest another potential mechanism for surfactant dysfunction within the injured lung. Alterations in the cholesterol-to-phospholipids ratio of surfactant alter surfactant function. As noted earlier, cholesterol is the major neutral lipid in surfactant and when present in physiologic amounts, it enhances adsorption at the air–liquid interface. Studies focused on the specific role of neutral lipids and cholesterol in lung injury are scarce. *In vitro* studies that have altered the cholesterol-to-phospholipid ratio show that supraphysiologic levels of cholesterol affect surfactant film stability leading to the inhibition of its function.⁸⁷ This cholesterol-mediated dysfunction appears to be relevant *in vivo* as well, because surfactant isolated from animals exposed to high tidal-volume ventilation showed abnormal biophysical activity, which was subsequently restored upon cholesterol removal.⁸⁸

Interestingly, these experimental results are consistent with clinical observations. Markart et al showed that patients with ARDS had increased levels of the neutral lipid component in isolated LA forms, and impaired

biophysical activity of their endogenous surfactant.⁸⁹ The source of the increased cholesterol in the surfactant of injured lungs is currently unknown. Altered synthesis by alveolar epithelial type II cells and/or leakage from the circulation into the alveolus are potential mechanisms, although this has not been formally evaluated. Further studies are necessary to address these issues, as well as to understand the biophysical mechanisms leading to cholesterol-mediated inactivation of surfactant.

Finally, within a permeable lung, characteristic of patients with ARDS, serum proteins such as albumin, hemoglobin, and fibrinogen can leak into the airspace and competitively inhibit the function of remaining LA forms.⁹⁰ This latter mechanism has been extensively studied over the past several years, and it has been shown that this inhibitory phenomenon may be overcome via the administration of large amounts of an effective exogenous surfactant preparation, thus providing important rationale for this therapeutic approach.⁹¹

Exogenous Surfactant Administration in the Injured Lung

Results of Phase II and Phase III Clinical Trials

Exogenous surfactant administration has been evaluated in patients with ARDS based on the rationale that surfactant alterations contribute, at least in part, to the lung dysfunction associated with this disorder. Moreover, preclinical studies evaluating the efficacy of exogenous surfactant administration in animal models of ALI have shown promising results. Unfortunately, the outcomes of clinical trials conducted to date have been variable (Table 60-1). For example, the first large, randomized phase III clinical trial, involving more than 700 patients, administered aerosolized Exosurf (a preparation with no surfactant proteins) to patients with severe, sepsis-induced ARDS over a 5-day period.⁹² Mortality was similar in the treatment (41%) and standard care (41%) groups. A subsequent, albeit smaller trial, with approximately fifty patients, evaluated tracheally instilled Surfactant (a natural bovine preparation) in a relatively similar patient population with good results: a mortality of 19% compared to 44% in the control group.⁹³ In this latter study, the surfactant preparation contained both SP-B and SP-C, and much larger quantities of the material reached the airspace compared to the Exosurf trial. Other, relatively small clinical studies have included a phase II trial that used a porcine natural surfactant preparation, HL-10, which also contained SP-B and SP-C. The surfactant was instilled in large doses into the lungs of patients with severe ALI or ARDS. After somewhat promising results, a much larger, randomized phase III clinical trial was conducted, but was stopped prematurely for lack of efficacy.⁹⁴

Table 60-1: Published Phase II and Phase III Clinical Trials Involving Exogenous Surfactant

Surfactant Preparation and Concentration	Delivery Method	Dose	Mortality (vs. Standard Therapy)	Ref.
Exosurf (13.5 mg DPPC/mL)	Aerosolization	112 mg/kg/day × 5 days	41% vs. 41%	92
Survanta (25 mg DPPC/mL)	Bolus instillation	4 or 8 doses of 50 or 100 mg/kg	19% vs. 44%	93
Venticute (25 mg DPPC/mL)	Bolus instillation	Up to 4 doses of 50 mg lipid/kg	36% vs. 32%	96

Surfaxin, a synthetic surfactant preparation containing an SP-B–like peptide (KL-4), was tested in twelve patients with severe ARDS and administered via a lung-lavage procedure.^{[95](#)} Although these initial results were promising, there has been nothing further reported regarding results of a larger study.

Results from a large, randomized phase III trial that tested the efficacy of an exogenous surfactant preparation composed of a recombinant SP-C protein with lipids (Venticute) have been published.^{[96](#)} The material was instilled into the lungs of patients with severe ARDS caused by multiple etiologies, and results showed no overall difference in mortality between the surfactant-treated group (36%) and standard therapy (32%). A post hoc analysis of these results however, showed a significant interaction between the surfactant treatment and the mechanism of lung injury. Patients with “direct” lung injuries, induced by pneumonia and/or aspiration, had a statistically significantly higher survival rate when treated with surfactant compared to standard therapy; no such relationship existed for patients with “indirect” causes of ARDS, such as sepsis and trauma.

This analysis resulted in a follow-up, prospective, randomized, blinded phase III clinical trial focused on the administration of the recombinant SP-C–based surfactant in patients with severe, direct lung injuries. Surprisingly, the results showed that exogenous surfactant instillation did not improve mortality, oxygenation, or the incidence of nonpulmonary organ failures and the study was stopped for futility.^{[97](#)} This unexpected lack of efficacy was unexplained, but was thought to be related, in part, to a possible inactivation of the material during the reconstitution procedure.

Although results of clinical trials are inconsistent and certainly disappointing, the information gained from these trials, as well as preclinical studies, is insightful and supports the need for further research in this area. In the next section, the various factors that influence a host’s response to exogenous surfactant are reviewed.

Factors Influencing the Efficacy of Exogenous Surfactant

Surfactant Preparation

A number of surfactant preparations have been tested in clinical trials involving patients with ARDS, and are generally classified based on their surfactant protein content ([Table 60-2](#)). Natural surfactant products contain various amounts of SP-B and SP-C along with natural lipids. They are derived from either bovine (Survanta, Infasurf, Alveofact, and [bovine lipid extract surfactant](#) [BLES]) or porcine (Curosurf and HL-10) sources, and have consistently been shown to have excellent biophysical activity. Currently available synthetic surfactant preparations contain either synthetic or recombinant surfactant-specific proteins combined with commercially available lipids. These preparations include Venticute and Surfaxin. Venticute is composed of a recombinant SP-C protein together with DPPC and smaller amounts of palmitoylphosphatidyl glycerol (POPG) and palmitic acid.⁹⁶ Surfaxin contains a synthetic SP-B-like peptide as well as DPPC, POPG, and palmitic acid.⁹⁵ Both Venticute and Surfaxin have been tested in preclinical studies involving models of ALI and phase II and phase III clinical trials. Two other synthetic surfactant preparations previously used in neonates with NRDS and tested in patients with ARDS include Exosurf (DPPC; hexadecanol, and tyloxapol) and artificial lung-expanding compound (ALEC, DPPC, and phosphatidylglycerol).⁹⁸ These preparations contain no surfactant proteins, perhaps explaining the poor results of these latter trials. No further studies are being conducted with these preparations at this time. Animal studies that have directly compared the various surfactant preparations have generally shown that those containing surfactant proteins, either from natural or synthetic or recombinant sources, are superior to those having no proteins.^{99,100} Furthermore, meta-analyses of several studies involving neonates with NRDS also show that clinical trials utilizing natural products result in better outcomes than those using the synthetic preparations with no surfactant proteins.¹⁰¹

Table 60-2: Exogenous Surfactant Preparations

Brand Name	Source/Composition
<i>Natural Protein-Containing</i>	
Survanta	Bovine-lung tissue extract
BLES	Bovine-lung lavage extract
Infasurf	Bovine-calf lavage extract
Alveofact	Bovine-lung lavage extract
Curosurf	Porcine-lung tissue extract
HL-10	Porcine-lung tissue extract
<i>Synthetic/Recombinant Protein-Containing</i>	
Surfaxin	DPPC, POPG, PA, “SP-B-like peptide”
Venticute	DPPC, POPG, PA, rSP-C
<i>Non-Protein-Containing</i>	
ALEC	DPPC, PG
Exosurf	DPPC, hexadecanol, tyloxapol

Abbreviations: ALEC, artificial lung-expanding compound; BLES, [bovine lipid extract surfactant](#); DPPC, dipalmitoylphosphatidylcholine; PA, palmitic acid; PG, phosphatidyl glycerol; POPG, palmitoyloleoylphosphatidyl glycerol; rSP-C, recombinant human SP-C.

Based on preclinical data and results of these clinical trials, it is likely that some type of protein-containing surfactant will be required for optimal results in patients with ALI or ARDS. Whether the natural products will be more effective than the protein-containing, synthetic surfactants in this setting is unknown, and will be difficult to prove in this complex patient population. As a result, other factors may need to be considered when making this decision. For example, although animal-based products may well be as effective or superior to any synthetic products available (or to be developed in the future), they have the theoretical potential of transmitting molecules and/or infectious agents to the host. This, however, has not been demonstrated over the past 20 years of use in neonates. In addition, given the potentially large patient

population that would ultimately benefit from this therapy, even if natural products are shown to be superior, availability may be a factor because of resource limitations. These issues do not exist for the recombinant and synthetic preparations, perhaps favoring further development of effective and easily manufactured synthetic products.

Surfactant Delivery and Dosing

The various delivery methods that have been utilized for administering exogenous surfactant to patients with ALI include: (a) instillation of a liquid bolus through the endotracheal tube,⁹⁶ (b) bronchoscopic instillation of smaller aliquots of surfactant to the various segments of the lung,¹⁰² (c) sequential segmental lavage of lung units with a surfactant preparation via the bronchoscope,⁹⁵ and (d) aerosolization of surfactant via a nebulizer, either in liquid form or as a dry powder.^{92,103} For all methods of delivery, the common goal is to deliver sufficient amounts of material to the distal lung in an optimal distribution pattern. Unfortunately, the lung injury in critically ill patients is often non-uniform and varies over time as lung dysfunction deteriorates. As a result, it is likely that individual treatment strategies may have to be tailored to the specific patient involved. For example, patients with severe lung dysfunction with a nonuniform distribution of injury would not benefit from any aerosolized surfactant preparation delivered to the whole lung, because relatively small quantities of surfactant are deposited within lung tissue and most of the material would be deposited in the most compliant regions of the lung, areas that need it the least.¹⁰⁴ In this situation, delivering large amounts of surfactant as a liquid bolus (50 to 200 mg lipid/kg body weight), either through the endotracheal tube or in sequential aliquots (with or without segmental lavage) via the bronchoscope, may be superior. The lavage approach may offer the advantage of removing inflammatory cell products and serum proteins, which tend to inhibit surfactant function, while leaving sufficient quantities of surfactant behind in the alveoli to improve lung function.^{95,105} Although this latter approach is rather invasive, more time-consuming, and requires a skilled bronchoscopist to perform the procedure when compared to the bolus instillation technique, it may be optimal for specific types of severely ill patients. Currently, there is little doubt that patients with severe lung injury will require relatively large doses of surfactant to overcome surface tension inhibitory forces within the injured lung, the latter being induced either by serum proteins leaking into the airspace or inactivation of surfactant via nitration or oxidation or cholesterol, as previously described.^{55,80,81,87,88,90}

Aerosolization would likely only be applicable to patients with milder disease who require smaller amounts of material and have relatively uniform injuries, a situation in which exogenous surfactant administration has yet to be tested in the clinical setting. In addition, a consistent and cost-effective nebulizer able to deliver reasonable amounts of surfactant over an extended period of time is not yet available, although dry powder delivery devices may achieve these objectives in the future.

Despite the numerous studies that have evaluated these various delivery methods in animals and humans, it is still not clear which delivery method should be used for specific types of patients, how much surfactant should be delivered per dose for the various delivery methods available, and how many doses will be

required over time for each patient to achieve an optimal outcome. More research is certainly required in this area.

Effects of Mechanical Ventilation

Mechanical ventilation is an important supportive therapy for patients with ALI or ARDS, and has recently been shown to impact the outcome of these patients.⁸⁶ Similar to its impact on the endogenous surfactant system, the specific mode of mechanical ventilation used in surfactant-treated lungs can influence the metabolism of the administered material. For example, animal studies have shown that lower tidal volumes resulted in superior physiologic responses compared to higher tidal volumes after exogenous surfactant was delivered as a liquid bolus; this was associated with less conversion of the administered LA forms into dysfunctional SA.^{106,107} Clinically, these findings suggest that the mode of mechanical ventilation may have a significant impact on the duration of response to a particular dose of surfactant, which, in turn, would decrease the number of doses required. It is also important to optimize ventilation during and immediately after surfactant administration in order to optimize the distribution of the material and prevent airway obstruction. Positive end-expiratory pressure levels must be maintained during instillation to prevent alveolar collapse, and adequate tidal volumes must be delivered immediately after instillation to maintain recruitment and enhance peripheral distribution of the material.^{106–108}

Similar to the situation in the non-surfactant-treated lung, the use of recruitment maneuvers in conjunction with exogenous surfactant is somewhat controversial, with both beneficial and even harmful effects reported.^{109,110} It is likely that the timing and specific nature of the maneuver (i.e., frequency, duration, etc.) will impact its effects in the clinical setting of surfactant administration.

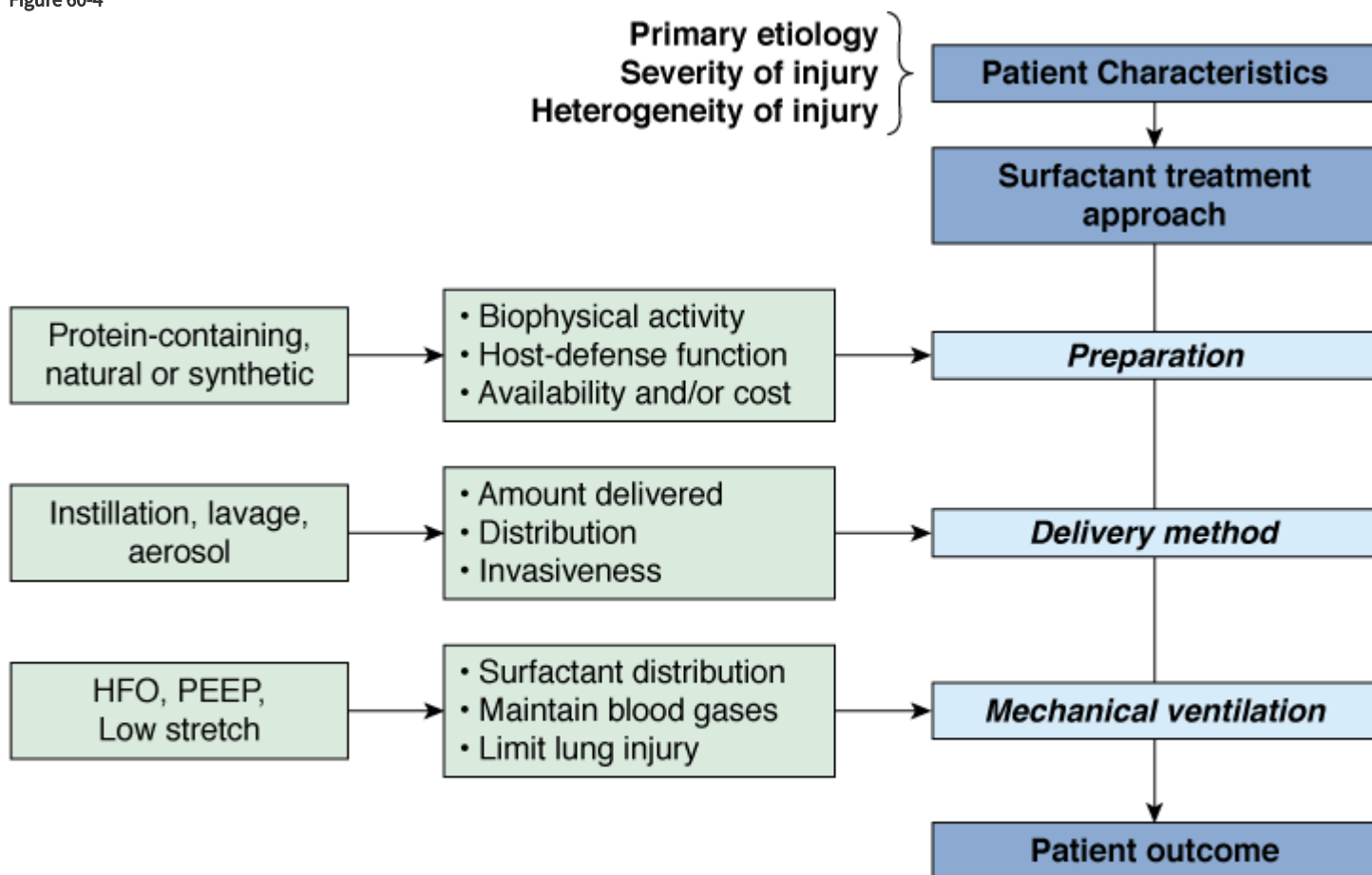
Finally, given the promising, albeit early, results of HFO in patients with ALI, some comments regarding its potential in the setting of exogenous surfactant administration are warranted. Very few studies have been performed to date, and results are inconsistent. Froese et al demonstrated that HFO was superior to conventional modes of ventilation after surfactant administration in saline-lavaged adult rabbits, but relatively high mean airway pressures were required to maintain lung recruitment with HFO.¹¹¹ On the other hand, a study in saline-lavaged adult sheep showed that exogenous surfactant may obstruct conducting airways when delivered in association with HFO, and that a period of more conventional tidal ventilation during and immediately following administration may be required for optimal distribution before switching over to HFO.¹¹² Further studies are required to determine how HFO and recruitment maneuvers should be used in conjunction with exogenous surfactant administration in patients with ALI or ARDS.

Nature of the Underlying Injury

As alluded to in previous sections, the efficacy of a particular surfactant treatment strategy may predominantly depend on the specific type of patient involved, and, in this regard, influence all of the factors previously described (Fig. 60-4). ALI and ARDS can be initiated by several different insults, including those

directly affecting the lung, such as aspiration and pneumonia, as well as indirect causes, such as systemic sepsis and trauma. The results of a recent phase III clinical trial, involving the synthetic surfactant preparation Venticute administered to patients with various types of ARDS, suggested that patients with direct lung injuries initiating the development of the disease would benefit the most from surfactant therapy.⁹⁶ A follow-up study, however, specifically focused on patients with direct lung injuries failed to confirm those observations, although there was some question of inactivation of the surfactant before administration.⁹⁷

Figure 60-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Various factors may influence a host's response to exogenous surfactant, including specific surfactant preparation, dosing and delivery method, mode of mechanical ventilation implemented during and after surfactant administration, and nature of the underlying injury. See text for details. *HFO*, high-frequency oscillation; *PEEP*, positive end-expiratory pressure.

Previous studies have shown that several host factors, as well as timing of the intervention may influence a patient's response to surfactant; thus predicting consistent outcomes can be a problem in this patient population. Unfortunately, animal models that accurately reflect the various types of patients who develop ARDS, and ultimately those who would respond to exogenous surfactant, are lacking. Patients with ARDS have a complex pathophysiology involving various types and severities of insults. As a result, we are left with the challenge of pursuing large, expensive, and time-consuming clinical trials with no guarantees that they

will demonstrate beneficial results. It is imperative, therefore, that we gain as much insight as possible from the preclinical and clinical studies that have been conducted to date

Future Research Directions

Combination Therapies

The biophysical properties of surfactant make it a suitable candidate for combination therapies involving other agents.¹¹³ For example, an effective surfactant adsorbs readily to the air–liquid interface and spreads rapidly across the alveolar space.^{3,4,41} This puts the surfactant, as well as the added agent, in close proximity to alveolar epithelial cells as well as the pulmonary vasculature, ideally with a good distribution pattern. These particular characteristics would result in high local pulmonary concentrations of a therapeutic agent with minimal side effects or toxicity. Indeed, surfactant enhances the delivery as well as the expression of an adenoviral vector, and improves the peripheral distribution and activity of recombinant superoxide dismutases and antibiotics.^{114–116} The combination of exogenous surfactant with other physiologically active compounds, such as [nitric oxide](#) and perfluorocarbons, also has been tested and shown to have synergistic effects.^{117,118} Future applications for surfactant as a carrier vehicle and/or as part of a cocktail mixture for patients with lung injury require further investigation.

Role of Surfactant in Other Diseases

The vast majority of surfactant research that has focused on the mature lung has involved animal models and/or patients with ALI or ARDS. Recent evidence suggests, however, that alterations in the endogenous surfactant system also may be implicated in several other diseases of the lung, which potentially could lead to novel therapeutic approaches to these disorders in the future. One such disease reflecting alterations in endogenous surfactant metabolism is pulmonary alveolar proteinosis.¹¹⁹ Increased synthesis and secretion together with decreased catabolism of surfactant results in excessive accumulation of surfactant lipids and proteins within the airspace. Removal of this excess material via whole-lung lavage has resulted in long-term benefits for most patients.

In addition, given the extensive *in vitro* and *in vivo* data showing that the various components of surfactant can affect immune-cell regulation, inflammatory cell responses, and bacterial and viral proliferation/killing, it is likely that surfactant also plays an important role in bacterial and viral pneumonia. Surfactant alterations have been consistently reported in these patients. A clinical trial involving children with respiratory syncytial virus showed improved lung function and shorter hospital stays in those receiving exogenous surfactant compared to non–surfactant-treated control subjects.^{120,121} It is difficult, however, to separate the biophysical aspects of the improved lung function observed from the immunomodulatory effects of surfactant on the development and progression of the infection itself. More animal studies and clinical trials evaluating early exogenous surfactant administration in this setting are necessary to determine if surfactant plays an important role in pulmonary infections.

Surfactant alterations also have been documented in patients with various interstitial lung diseases. These reports have shown decreased lipid and SP-A levels in the bronchoalveolar lavage fluid of patients with idiopathic pulmonary fibrosis, which in one study predicted 5-year survival.^{122,123} SP-A levels were actually increased in the lungs of patients with sarcoidosis and hypersensitivity pneumonitis, suggesting that, similarly to the various types of ARDS, the role of surfactant may vary according to the specific type of interstitial disease and insult involved.¹²⁴

Various studies suggest that surfactant plays an important role in obstructive lung diseases.¹²⁵ Surfactant facilitates ciliary function, downregulates some of the inflammatory processes implicated in these diseases (asthma and cystic fibrosis), and is important in maintaining the patency of conducting airways.^{50,51} Although various studies report surfactant alterations in most obstructive diseases, only a few small studies have tested the efficacy of exogenous surfactant administration in this setting.¹²⁶ The results of these preliminary studies suggest that surfactant may be effective as therapy in asthma, although further research is required.

Finally, surfactant changes were demonstrated in the ischemia–reperfusion injury associated with lung transplantation. Exogenous surfactant, administered to the donor lung before storage, mitigated this injury, even in the setting of prolonged organ storage.¹²⁷ This indication for surfactant administration has not been tested in clinical trials to date.

Summary and Conclusions

Pulmonary surfactant is critical for maintaining alveolar stability and normal lung function. Although the composition of surfactant is relatively consistent among mammalian species, there is still much to be learned regarding the specific functions of the individual components of surfactant, particularly with respect to host defense. Most research to date involving surfactant perturbations in the mature lung has focused on the lung injury associated with ALI or ARDS, and has led to clinical trials evaluating the efficacy of exogenous surfactant administration in this setting. Unfortunately, results of these trials have been inconsistent, and somewhat disappointing. Careful analyses of these results, together with extensive preclinical studies, have provided important insight into the various factors that may influence a host's response to this therapy. Research conducted over the next few years should take advantage of the lessons learned from previous studies, and focus on the most important factors influencing a host's response to this therapy in various clinical settings, thereby optimizing patient outcomes.

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